

Learned Image Compression with Enhanced Dynamic Spatial Aggregation and Asymmetric Entropy model

Yuanton Zhang^{†1}, Nianxiang Fu^{†2}, Guo Yu³, Xiangdong Lv³, Huairui Wang¹, Zhenzhong Chen^{1,2,3}

¹*School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China*

²*Hubei Luojia Laboratory, Wuhan University, Wuhan, China*

³*School of Computer Science, Wuhan University, Wuhan, China*

Abstract—Learned image compression (LIC) has shown significant potential and better rate-distortion performance than traditional techniques. However, existing CNN-based approaches or window-based self-attention methods can only capture spatial information within fixed ranges. To tackle this limitation, we propose a novel method with dynamic spatial aggregation for transform coding. Our approach introduces enhanced adaptive aggregation, which generates kernel offsets to capture relevant information within content-dependent ranges, improving the transform process. Furthermore, we define a generalized coarse-to-fine entropy model that considers global context, channel-wise information, and spatial context in a coarse-to-fine manner. Additionally, our method takes into full consideration the model's efficiency and complexity. By introducing the asymmetric entropy model structure and efficient heterogeneous convolution, our approach maintains lower coding complexity and higher decoding speed while ensuring performance. Experimental results demonstrate a substantial improvement in rate-distortion performance achieved by our method when compared to traditional compression methods like VTM and BPG, as well as some LIC methods.

Index Terms—learned image compression, dynamic kernel, adaptive spatial aggregation, efficient entropy model.

I. INTRODUCTION

IN recent years, learned image compression (LIC) has revolutionized image compression. While traditional methods like JPEG[1] and VVC[2] have been the standard for visual data compression, the rapid progress in deep learning is leading us into a new era with data-driven approaches to overcome traditional limitations.

While some recent learning-based image compression methods[3], [4], [5], [6] have indeed surpassed VTM in terms of performance. Many of these methods either rely on Convolutional Neural Networks (CNN) [7], [8], [9] or employ window-based attention mechanisms[10], [3] to aggregate image content information. Consequently, these models have fixed receptive fields and lack the adaptability to dynamically aggregate information based on specific image content. In this paper, we focus on enhancing the spatial aggregation capacity in a dynamic range and reducing decoding complexity, aiming

to improve the rate-distortion performance of the model further while maintaining reasonable coding speed. In particular, we introduce Lite DCN[11], which can generate learned offsets and group-sharing weights to break the limitation of fixed-range spatial aggregation and have more flexible receptive fields. In order to further reduce computational complexity and improve inference speed, we also introduce heterogeneous convolutions[12] in the feature transformation structure. As another core component of LIC, entropy models play a crucial role in parameter estimation and probabilistic prediction. Balle et al. [8] introduce a hyper-prior for global context, while Minnen et al. [13] employ a spatial autoregressive model for context. However, this context model lacks parallel processing due to its raster scan order. To address this, Minnen et al. [14] propose a channel-wise autoregressive entropy model by slicing the latent along the channel dimension for sequential encoding. He et al. [15] propose a checkerboard context model, dividing the latent into two spatial parts for dual-pass encoding. Furthermore, He et al. [5] combine the checkerboard context model with unevenly grouped channel-conditional context. In this paper, we first define a generalized efficient entropy model using a coarse-to-fine context. Then, we introduce the abovementioned dynamic kernel in hyper-prior to generate a more expressive global context. Moreover, we utilize the energy reduction and the spatial correlation in the latent and propose the Asymmetric Spatial-Channel Context (ASCC) Entropy Model to promote RD performance while keeping a satisfying decoding speed.

II. METHODS

A. Overall framework

The proposed scheme is based on our previous work[16], the compression framework can be formulated as follows:

$$\begin{aligned} \mathbf{y} &= g_a(\mathbf{x}; \phi) \\ \hat{\mathbf{y}} &= Q(\mathbf{y} - \mu) + \mu \\ \hat{\mathbf{x}} &= g_s(\hat{\mathbf{y}}; \theta) \end{aligned} \quad (1)$$

where $\mathbf{x}$ and $\hat{\mathbf{x}}$ denote the input image and its decompressed counterpart. g_a and g_s represent the analysis and synthesis transforms. ϕ and θ are parameters of the analysis and the

[†] denotes equal contribution. Corresponding author: Zhenzhong Chen.(Email: zzchen@whu.edu.cn) This work was supported by the National Natural Science Foundation of China under contract No. 62036005 and Tencent.

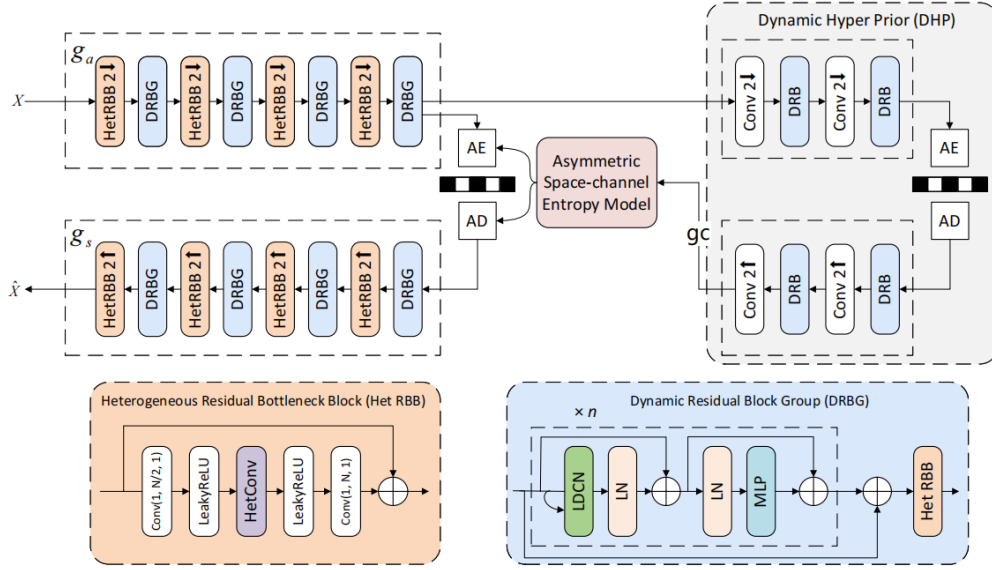


Fig. 1. The framework of the proposed image compression framework. We use g_a to transform the image (x) into latent representation y , and propose an asymmetric space-channel entropy model to estimate the parameters (μ, σ) of Gaussian distribution.

synthesis transform module learned with neutral networks. Q means quantization operation. During encoding, the input x is sent to the encoder to obtain the latent representation y . We follow previous work [14] to encode the residual between y and μ into the bitstream. The quantized residual $Q(y - \mu)$ is encoded using arithmetic coding with quantization Q . In the decoding process, the decoder g_s transforms the decoded latent $\hat{y}$ into the reconstructed image. We model symbols using a single Gaussian distribution based on previous studies [8], [13]. Specifically, the means μ and variances σ of the symbols are estimated using the entropy model. To construct a comprehensive entropy model in a step-by-step manner, we incorporate the hyper-prior [8], channel-wise autoregressive context [14] and spatial context entropy model [15], [5] into the framework. The generalized entropy model encodes y into side information z using the hyper-prior network. Then, y is divided into N slices ($y^1, y^2, \dots, y^N$) along the channel dimension, and each slice is encoded in an autoregressive manner. In the parameter estimation of each slice, the previously decoded slices assist in estimating the distribution of the current slice. Additionally, each slice (e.g., the i -th slice) is further divided into K parts in the spatial dimension ($y_1^i, y_2^i, \dots, y_K^i$). The complete formulation of the entropy model is as follows:

$$\begin{aligned}
 z &= h_a(y; \phi_h), \hat{z} = Q(z) \\
 gc &= h_s(\hat{z}; \theta_h) \\
 cc &= g_{cp}(\hat{z}, \hat{y}^{<i}; \theta_{cp}), 1 \leq i \leq N \\
 sc &= g_{sp}(\hat{z}, \hat{y}_{<j}^i; \theta_{sp}), 1 \leq j \leq K \\
 \mu_j^i, \sigma_j^i &= g_{ep}(gc, cc, sc; \theta_{ep}), \\
 p_{\hat{y}}(\hat{y}_j^i | ctx) &= [\mathcal{N}(\mu_j^i, (\sigma_j^i)^2)] * \mathcal{U}(-\frac{1}{2}, \frac{1}{2})(\hat{y}_j^i) \\
 \text{with } ctx &= (\hat{z}, \hat{y}^{<i}, \hat{y}_{<j}^i, \theta_h, \theta_{cc}, \theta_{sc}, \theta_{ep})
 \end{aligned} \tag{2}$$

In this framework, the terms h_a , h_s , ϕ_h , and θ_h refer to the hyper analysis and hyper synthesis transforms, along with their respective learnable parameters within the hyper-prior. The variables z and $\hat{z}$ represent the side information and its quantized counterpart. The networks g_{cp} , g_{sp} , g_{ep} , ϕ_{cp} , ϕ_{sp} , and θ_{ep} correspond to the channel, spatial, and final parameter prediction networks, and their associated learnable parameters. By feeding $\hat{z}$ into the hyper synthesis transform h_s , we obtain the global coarse context denoted as gc . Furthermore, utilizing gc and the previously decoded slices ($y^1, y^2, \dots, y^{i-1}$), we generate the channel-wise context cc using the network $g_{cp}(\cdot)$. Simultaneously, the entropy model generates the spatial context sc by utilizing the decoded parts ($y_1^i, y_2^i, \dots, y_{j-1}^i$). Finally, the Gaussian parameters μ_j^i and σ_j^i for the target position are predicted by the final parameter prediction network g_{ep} (referred to as the "Entropy Parameters" network in [13]).

B. Dynamic Kernel for feature extraction

In earlier LIC methods [13], [9], [15], [5], CNNs were commonly used as feature extractors to obtain the latent representation of images. In recent years, with the success of transformer architectures in the field of computer vision, some end-to-end compression methods based on self-attention have also emerged [17], [3], [18]. However, methods based on convolutional neural networks are limited by the receptive field size of the convolutional kernels, and the translational invariance of convolutions results in fixed kernel weights when processing different content. Additionally, transformers-based methods are also limited by the window size. To overcome the limitations mentioned above and enhance the RD performance, we propose to utilize deformable kernels [19], [20] for adaptive spatial aggregation. By incorporating content-adaptive kernel offsets and weights in deformable convolution (DCN), the

kernel can dynamically aggregate spatial information across a wide range, facilitating efficient Gaussianization. Specifically, here we adopt Lite DCN (LDCN) [11] as the dynamic kernel to aggregate information in a deformable spatial range. The input feature in LDCN is divided into groups, sharing modulated kernel weights within each group. Additionally, LDCN employs softmax normalization instead of element-wise sigmoid normalization, which helps alleviate unstable gradients during training. The LDCN used in our method can be formulated as:

$$\hat{F}(\mathbf{p}) = \sum_{g=1}^G \sum_{k=1}^K w_{gk} \cdot F(\mathbf{p} + \mathbf{p}_k + \Delta \mathbf{p}_{gk}), \quad (3)$$

where G denotes the number of partition groups, w_{gk} and $\Delta \mathbf{p}_{gk}$ represent the shared modulated kernel weights and the learned offsets within the g -th sub-group, respectively. Thanks to the group-wise sharing weights, the Lite DCN utilized in our approach can be seamlessly integrated as a dynamic transformation operation without imposing significant computational overhead. Taking inspiration from recent advancements in Vision Transformers [21], [22], we have also integrated Layer Normalization[23] (LN) and Multilayer Perceptron (MLP) to enhance the network's modeling capability. In particular, as illustrated in Fig. 1, we apply Layer Normalization (LN) after the LDCN and MLP structures. Additionally, we incorporate shortcut connections [24] into both the LDCN and MLP, aligning with the design principles of recent studies [21], [25], [22], forming what we refer to as the Dynamic Residual Block (DRB). To mitigate computational complexity, we employ heterogeneous convolutions[12] (HetConv) within the residual bottleneck block, which constitutes Heterogeneous Residual Bottleneck Block (Het RBB). HetConv reduces computational requirements (FLOPs) and model parameters compared to standard convolution operations while preserving representational efficiency. Moreover, we introduce the Dynamic Residual Block Group (DRBG), which serves as the primary component of the transformation network, as depicted in Figure 1. In summary, we utilize adaptive spatial aggregation in a dynamic range for efficient Gaussianization to design our analysis and synthesis transforms. Furthermore, thanks to the utilization of grouped convolution and heterogeneous convolution, our model maintains low computational complexity while ensuring performance.

C. Asymmetric Spatial-Channel Context Entropy Model

To facilitate the estimation of distribution parameters, recent advancements in efficient entropy models utilize diverse contextual information [8], [14], [15]. In this study, to utilize all these previous methods' advantages, we divide the helpful context in image compression into coarse global, channelwise, and spatial contexts.

As Equation 2 described, we obtain the coarse global context, using a dynamic hyper-prior, as side information, which has been shown effective in previous studies [13], [9]. The architecture of the Dynamic Hyper Prior is shown in Figure 1. For the fine-grained context, following the approach in [14],

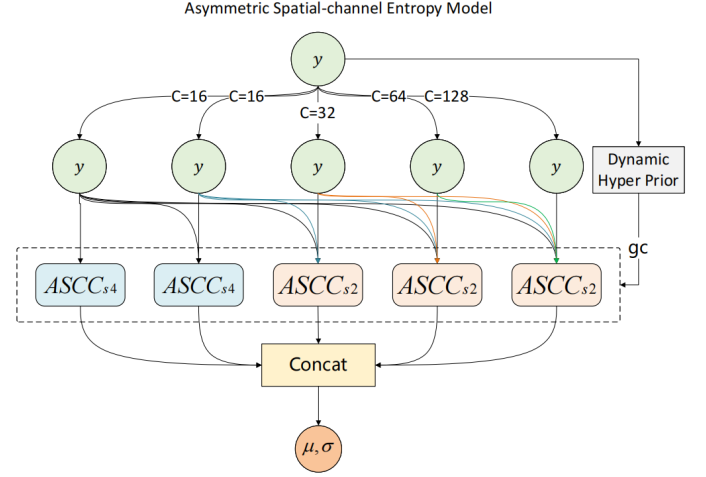


Fig. 2. The framework of the proposed asymmetric spatial-channel context entropy Model. The latent y is split into 5 slices, each slice utilizes hyper-priors to help generate Gaussian parameters. Besides, considering that different slices contain varying amounts of information, we use a 4-stage checkerboard model for the first two slices to estimate Gaussian parameters, while a coarser 2-stage checkerboard model is used for the subsequent slices.

[15], we split the latent representation along the channel and spatial dimensions. Considering the orthogonality of space and channel dimensions, the performance improvement from these two dimensions should also be orthogonal. Specifically, we create unevenly grouped channel-wise context, as the later encoded groups contain less information [5]. We set the number of groups to 5, with 16, 16, 32, 64, and 192 channels in each group. Furthermore, due to the uneven distribution of information entropy among different groups, and the spatial correlation of image features gradually decreases during the process of channel auto-regression. This indicates that we can capture the more helpful spatial context in the former groups, and later channel groups are more dependent on the channel context than the spatial context. Therefore, as shown in Fig. 2 the space context model is also applied in an asymmetric manner. We utilize the four-stage spatial context model $ASCC_{s4}$ (inspired by the multi-stage context in [26]) to reduce spatial redundancy in the former channel group, and the two-stage spatial context model $ASCC_{s2}$ (inspired by the checkerboard context in [15]) is applied to the latter channel group. The entire model, named the Asymmetric Spatial-Channel Context (ASCC) Entropy Model, can mine valuable context in a comprehensive manner while maintaining a high inference speed.

III. EXPERIMENTS

1) *Datasets*: Building upon the previous research[27], we utilize the Flickr2W dataset [28] as training data. This dataset comprises 20,716 real-world images which have diverse and intricate textures. To assess the effectiveness of our model and compare its performance against other image compression methods, we conduct evaluations on three benchmark datasets: the Kodak dataset [29], the CLIC Professional Validation dataset [30], and the Tecnick dataset [31]. These datasets

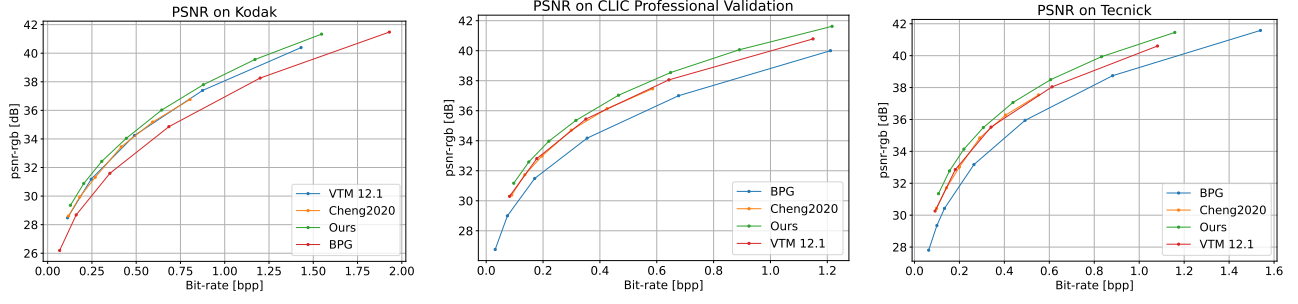


Fig. 3. The rate-distortion performance of BPG, VTM-12.1, Cheng2020 and our method on the Kodak, CLIC Professional Validation and Tecnick datasets.

TABLE I

THE COMPARISON RESULTS OF BDBR (ANCHOR: VTM-12.1), INFERENCE SPEED AND MODEL PARAMETERS. THE ENC-T AND DEC-T DENOTE THE TIME COST FOR ENCODING AND DECODING ON KODAK, AND PARA. REPRESENTS THE PARAMETERS OF EACH MODEL. **RED** INDICATE THE BEST PERFORMANCE.

Image Codec	Kodak	CLIC	Tecnick	EncT (ms)	DecT (ms)	Para. (M)
BPG444	17.78%	31.28%	23.40%	-	-	-
Cheng2020 [9]	1.34%	3.09%	0.02%	4250	10930	29.6
Ours	-8.08%	-10.33%	-10.97%	182	201	52.7

include images of various sizes, allowing us to evaluate our model's performance across different resolutions.

2) *Training settings*: To enable multi-batch parallel training, we randomly crop the training dataset to a size of 256×256 . To ensure that the model can simultaneously perceive the bitrate and reconstruction quality of the image, we follow the approach used in many current learned image compression methods [8], [9]. We incorporate a Lagrangian multiplier in the loss function to balance the trade-off between rate and distortion. Our framework is trained using the following loss function:

$$\begin{aligned}
 L &= \lambda \cdot D(\mathbf{x}, \hat{\mathbf{x}}) + R(\hat{\mathbf{y}}) + R(\hat{\mathbf{z}}), \\
 \text{with } R(\hat{\mathbf{y}}) &= \mathbb{E}[-\log_2(p_{\hat{\mathbf{y}}|\hat{\mathbf{z}}}(\hat{\mathbf{y}}|\hat{\mathbf{z}}))], \\
 R(\hat{\mathbf{z}}) &= \mathbb{E}[-\log_2(p_{\hat{\mathbf{z}}|\psi}(\hat{\mathbf{z}}|\psi))].
 \end{aligned} \quad (4)$$

The predicted entropy is denoted by $R(\cdot)$. The factorized entropy model ψ [7] is employed to compress the side information. The Lagrange multiplier λ determines the trade-off between the rate R and the distortion D . In our model training, we utilize Mean Squared Error (MSE) as the distortion function. To control the rate-distortion (RD) performance, we train eight models with different values of λ . Following the approach of CompressAI [32], we set the values of λ to 0.0018, 0.0035, 0.0067, 0.0130, 0.0250, 0.0483, 0.0932, and 0.1800 for each model. The batch size is set to 8, and we employ the AdamW optimizer with parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The learning rate is initialized to 1×10^{-4} and reduced to 1×10^{-5} after 380 epochs of training. The entire network converges after 400 epochs. All experiments are conducted using PyTorch on NVIDIA RTX 3090 GPUs.

3) *Comparison results*: We conducted a comparative analysis of our method against two traditional image compression methods BPG [33] and VTM-12.1 [34], as well as a learned image compression approach and a representative learned image compression approach Cheng2020[9]. Specifically, we obtain BPG from the website¹, and we set quantization parameter (QP) as 22, 27, 32, 37, 42, 47 for BPG. As for VTM-12.1, we download the reference software from the website². We follow [18] and use the scripts in CompressAI³ [32] to gather VTM results using QP 17, 22, 27, 32, 37, 42, 47.

Fig. 3 and Table I show the experimental results on the Kodak, CLIC Professional Validation, and Tecnick datasets while taking PSNR as quality measurement. Our method can outperform traditional image compression methods BPG and VTM-12.1 on every benchmarks. Specifically, the proposed method can save 8.08%, 10.33%, and 10.97% bitrate compared to VTM-12.1 on Kodak, CLIC, and Tecnick datasets, respectively. Compared to representative end-to-end image compression method Cheng2020, the proposed approach demonstrates significant performance advantages. Furthermore, thanks to introducing asymmetric entropy model structure and efficient heterogeneous convolutions, our model also outperforms Cheng2020 in terms of encoding and decoding speed.

IV. CONCLUSIONS

This paper introduces a novel learned image compression technique that incorporates dynamic kernel-based adaptive spatial aggregation and an asymmetric spatial channel entropy model. By leveraging dynamic kernels that generate content adaptive kernel offsets, our approach significantly enhances the transforming capacity through adaptive spatial aggregation. Furthermore, we introduce a generalized coarse-to-fine entropy model that considers global, channel-wise, and spatial contexts. To reduce statistical redundancy while maintaining high coding efficiency, we propose an asymmetric spatial channel entropy model and employ efficient heterogeneous convolution. Experimental results demonstrate the superior performance of our method compared to the traditional image compression method VTM-12.1 and other state-of-the-art approaches in terms of rate-distortion (RD) performance.

¹<https://bellard.org/bpg/>

²https://vcgit.hhi.fraunhofer.de/jvet/VVCSSoftware_VTM/-/tags/VTM-12.1

³<https://github.com/InterDigitalInc/CompressAI/tree/efc69ea24>

REFERENCES

- [1] G. K. Wallace, "The jpeg still picture compression standard," *Communications of the ACM*, vol. 34, no. 4, pp. 30–44, 1991. **1**
- [2] B. Bross, Y.-K. Wang, Y. Ye, S. Liu, J. Chen, G. J. Sullivan, and J.-R. Ohm, "Overview of the versatile video coding (vvc) standard and its applications," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 31, no. 10, pp. 3736–3764, 2021. **1**
- [3] R. Zou, C. Song, and Z. Zhang, "The devil is in the details: Window-based attention for image compression," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 17 492–17 501. **1, 2**
- [4] F. Chen, Y. Xu, and L. Wang, "Two-stage octave residual network for end-to-end image compression," in *Proceedings of AAAI conference on artificial intelligence*, 2022. **1**
- [5] D. He, Z. Yang, W. Peng, R. Ma, H. Qin, and Y. Wang, "Elic: Efficient learned image compression with unevenly grouped space-channel contextual adaptive coding," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. **1, 2, 3**
- [6] D. Wang, W. Yang, Y. Hu, and J. Liu, "Neural data-dependent transform for learned image compression," in *Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition*, 2022, pp. 17 379–17 388. **1**
- [7] J. Ballé, V. Laparra, and E. P. Simoncelli, "End-to-end optimized image compression," in *International Conference on Learning Representations*, 2016. **1, 4**
- [8] J. Ballé, D. Minnen, S. Singh, S. J. Hwang, and N. Johnston, "Variational image compression with a scale hyperprior," in *Proceedings of the International Conference on Learning Representations*, 2018. **1, 2, 3, 4**
- [9] Z. Cheng, H. Sun, M. Takeuchi, and J. Katto, "Learned image compression with discretized gaussian mixture likelihoods and attention modules," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 7939–7948. **1, 2, 3, 4**
- [10] Y. Zhu, Y. Yang, and T. Cohen, "Transformer-based transform coding," in *International Conference on Learning Representations*, 2021. **1**
- [11] W. Wang, J. Dai, Z. Chen, Z. Huang, Z. Li, X. Zhu, X. Hu, T. Lu, L. Lu, H. Li *et al.*, "Internimage: Exploring large-scale vision foundation models with deformable convolutions," *arXiv preprint arXiv:2211.05778*, 2022. **1, 3**
- [12] P. Singh, V. K. Verma, P. Rai, and V. P. Namboodiri, "Hetconv: Heterogeneous kernel-based convolutions for deep cnns," in *Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition*. Computer Vision Foundation / IEEE, 2019, pp. 4835–4844. **1, 3**
- [13] D. Minnen, J. Ballé, and G. D. Toderici, "Joint autoregressive and hierarchical priors for learned image compression," *Advances in Neural Information Processing Systems*, vol. 31, 2018. **1, 2, 3**
- [14] D. Minnen and S. Singh, "Channel-wise autoregressive entropy models for learned image compression," in *IEEE International Conference on Image Processing (ICIP)*. IEEE, 2020, pp. 3339–3343. **1, 2, 3**
- [15] D. He, Y. Zheng, B. Sun, Y. Wang, and H. Qin, "Checkerboard context model for efficient learned image compression," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 14 771–14 780. **1, 2, 3**
- [16] H. Wang, N. Fu, Z. Chen, and S. Liu, "Dynamic kernel-based adaptive spatial aggregation for learned image compression," *arXiv preprint arXiv:2308.08723*, 2023. **1**
- [17] M. Lu, P. Guo, H. Shi, C. Cao, and Z. Ma, "Transformer-based image compression," *arXiv preprint arXiv:2111.06707*, 2021. **2**
- [18] Y. Zhu, Y. Yang, and T. Cohen, "Transformer-based transform coding," in *International Conference on Learning Representations*, 2022. **2, 4**
- [19] J. Dai, H. Qi, Y. Xiong, Y. Li, G. Zhang, H. Hu, and Y. Wei, "Deformable convolutional networks," in *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 764–773. **2**
- [20] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable convnets v2: More deformable, better results," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2019, pp. 9308–9316. **2**
- [21] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10 012–10 022. **3**
- [22] Z. Liu, H. Hu, Y. Lin, Z. Yao, Z. Xie, Y. Wei, J. Ning, Y. Cao, Z. Zhang, L. Dong *et al.*, "Swin transformer v2: Scaling up capacity and resolution," in *Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition*, 2022, pp. 12 009–12 019. **3**
- [23] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016. **3**
- [24] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778. **3**
- [25] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M.-H. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in *Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition*, 2022, pp. 5728–5739. **3**
- [26] M. Lu and Z. Ma, "High-efficiency lossy image coding through adaptive neighborhood information aggregation," *arXiv preprint arXiv:2204.11448*, 2022. **3**
- [27] Y. Xie, K. L. Cheng, and Q. Chen, "Enhanced invertible encoding for learned image compression," in *Proceedings of the 29th ACM International Conference on Multimedia*, 2021, pp. 162–170. **3**
- [28] J. Liu, G. Lu, Z. Hu, and D. Xu, "A unified end-to-end framework for efficient deep image compression," *arXiv preprint arXiv:2002.03370*, 2020. **3**
- [29] E. Kodak, "Kodak lossless true color image suite (photocd pcd0992)," 1993. **3**
- [30] CLIC, "Workshop and challenge on learned image compression," in *Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition*, 2021. **3**
- [31] N. Asuni and A. Giachetti, "Testimages: a large-scale archive for testing visual devices and basic image processing algorithms," in *STAG*, 2014, pp. 63–70. **3**
- [32] J. Bégaint, F. Racapé, S. Feltman, and A. Pushparaja, "Compressai: a pytorch library and evaluation platform for end-to-end compression research," *arXiv preprint arXiv:2011.03029*, 2020. **4**
- [33] F. Bellard, "BPG image format," 2014, <https://bellard.org/bpg> Accessed: March 1, 2022. **4**
- [34] J. V. E. Team, "Vvc official test model vtm." 2021. **4**